EUREKA: Phase 3 - Key Insights

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***Abstract*—In this paper we present a high-level summary of our ADRDModel architecture and we show its generalization power across 6 international datasets with a wide range of features and population. We further describe more rigorous evaluation criteria to remove bias in ML performance experiments. Our methodology shows key design choices leading to a 21.7% improvement vs Phase2 models. Extensive analysis of contributing factors is shown for each of the studies, ranging from feature biases, correlations, biomarker, signal importance, short and long-term effects, geographic conditions, drug ingestion and MRI/PET imaging in early diagnosis.**

I. Phase-3 Model Architecture Overview

Our goal with Phase3 model is to provide a framework to study datasets beyond MHAS and the EUREKA challenge. In order to achieve this, we studied what ML components are actually generalizable enough to be problem-agnostic. This led to simplification of our Phase2 model, removing multiple of our models, as well as reducing data pre-processing that relies on data-assumptions (e.g. dataset-specific).

To abstract the ML concerns from medical practitioners, we built a framework to handle the entire end-to-end workflow, with built-in support for multiple datasets. We call this the Alzheimer's Disease and Related Dementias Model (ADRD-Model) package.

One of the main advantages is an inherited uncertainty estimation mechanism. ADRDModel provides multiple views of the same patient to build confidence on how different features sets can arrive to each conclusion.

Parameter optimization is already done for practitioners and validated over multiple datasets. We have found a set of parameters that balance explainability & performance by restricting model complexity when only marginal progress is done in performance. In Figure 1, we show ADRD architecture.

The Feature Fetch module provides an abstraction layer that has been optimize with feature-selection mechanism of pre-existing good features. This allows very fast integration between datasets, especially very large ones like The National Alzheimer's Coordinating Center (NACC) Uniform Data Set (UDS), during this module execution all known available useful features are extracted and engineered, all of this happens automatically for each supported dataset and if needed, it can be controlled in more detail by medical researchers.

Since ADRDModel has been generalized to operate over multiple studies, with no changes required, we instead defined a normalized study-agnostic dataset; this process is completed by the *DatasetAdapter*, which takes a study-specific data, like MHAS or NACC-UDS and harmonize it further following a common standard, this allow then an identical ARDRModel to be executed for any dataset, regardless of the study origin.

Another innovation from ADRDModel is the capability to provide a multi-output score, both indicating the expected cognitive score over a future-date as well as Health Score, a patient-specific score that estimates their risk considering their own factors, e.g. mild reduced cognition is expected with age, yet, it does not automatically mean out-of-range.

II. Performance Evaluation

The configuration of Phase2, being a single-study & a single-validation-set can lead to overly tailored design decisions that while they might provide improvement for Eureka challenge, it does not necessarily translate to multi-study benefits. During Phase2, if multiple sub-samples are taken for evaluation, we observed standard deviation as high as ± 1.28.

Our ADRD package provides built-in validation mechanism to test each hypothesis and verify if improvements are statistically significant. A subset of our experiments is reported in Table I, RMSE is shown for the continuous scores (cognitive strength), and AUC for the Health Score. Note that for datasets without z-normalization scores RMSE is not comparable.

The effects of overfitting during Phase2 can be observed when the same data is evaluated over multiple-samples (e.g. K-fold validation). During the live-competition, this team (NxGTR) achieved a score of 38.48 (RMSE), but a more rigorous validation shows 40.65, this aligns with prior observations that competition leaderboard results might be artificially better than reality might suggest. We believe a stricter validation mechanism (Table I) highlights better models even if the actual scores are lower.

III. Methodology

In this section we will explore how we achieved a 21.7% improvement in model performance compared to our original Phase2 approach, as well overall learnings from several international studies. By design, our ADRDModel will remain fixed, zero-changes over the course of these improvements, and all value will come from data preparation. The aim is to serve a better direction for medical practitioners to find actionable insights, rather than focus on ML algorithms tweaks.

Following this minimalist approach, our focus is on data. Although it is possible to marginally increase performance (<1%) with multiple model ensembles, e.g. Neural Network,

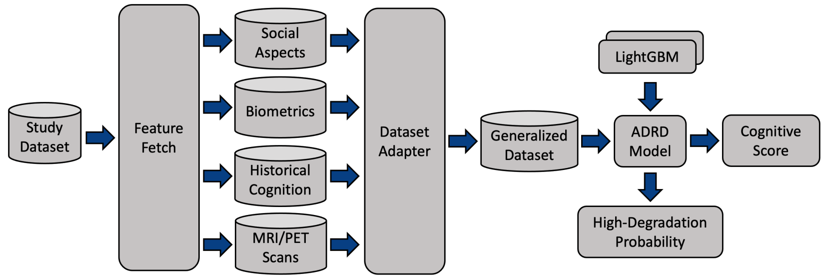

Fig. 1. Phase-3 Model Architecture Design

TABLE I
PHASE3 EXPERIMENTS SUMMARY

Dataset	Experiment	RMSE	AUC
Eureka	Phase2 Model	40.657 ±1.688	0.8807 ±0.023
MHAS	Phase2 Equivalent Extended Features	38.344 ±0.468	0.906 ±0.019
MHAS	With Biometrics	38.538 ±1.46	0.9059 ±0.02
MHAS	Biometrics & Historical Cognition	33.6134 ±2.005	0.9282 ±0.022
MHAS	Biometrics & Historical Cognition with Feature Selection	33.405 ±2.071	0.9285 ±0.022
HRS	Missing features removed	0.8454 ±0.025	0.7704 ±0.019
HRS	With Feature Selection	0.8438 ±0.024	0.7709 ±0.019
HRS	With Feature Selection & Education Bias Mitigation	0.8434 ±0.023	0.7732 ±0.018
MHAS	Transfer learning from HRS Model	33.579 ±2.177	0.9286 ±0.023
HRS	Transfer Learning From MHAS Model	0.8446 ±0.025	0.7717 ±0.02
LASI	With Feature Selection	0.6104 ±0.006	0.9066 ±0.003
CRELES	With Feature Selection	0.8346 ±0.055	0.8096 ±0.011
NACC	With Feature Selection	0.3797 ±0.008	0.7546 ±0.014
NACC	MRI Scan Data Only	0.6667 ±0.013	0.872 ±0.015
NACC	PET Scan Data Only	0.665 ±0.025	0.927 ±0.019

the loss in explainability and architecture complexity does not look worth it.

However, adding signals must be done with care as our collected datasets are not designed for ADRD but general usage, as a result most of the features are not relevant, Feature Selection is a necessity for explainable and consistent performance.

One must note that results across different competitors are not comparable unless subject to the same rigor standards, e.g. random K-fold validation might show lower performance than 1-single-set, yet it does not define a better model. In the next sections we review in detail the data used as well as top contributing factor for performance, since our approach is data-driven with a fixed general model to ensure it can operate even in yet-to-come datasets.

TABLE II
ADRDMODEL SUPPORTED DATASETS

Dataset	Reference	# Samples	# Features
DrivenData Eureka Challenge	Eureka [5]	2,002	183
Mexican Health & Aging Study	MHAS [6]	1,983	474 to 3384
United States Health and Retirement Study	HRS [7]	4,181	1090 to 1681
Longitudinal Aging Study in India	LASI [8]	35,363	461
Costa Rican Longevity and Healthy Aging Study	CRELES [9]	2,361	142
National Alzheimer's Coordinating Center	NACC [10]	8,991	305

IV. DATA COLLECTION

To ensure our design is adaptable to a variety of scenarios, we collected multiple studies datasets as shown in Table II.

We started by extending the original competition dataset to include the entire feature set available from MHAS, although later analysis shows our FeatureFetch module can reduce features without affecting performance.

The HRS dataset provides great historical tracking of waves to be able to evaluate different time-horizon prediction mechanisms, and comes with the largest set of features. Although based on our FeatureFetch module, we identified around 15% of features are responsible for most of the value, this is managed automatically for medical researchers.

The LASI dataset is by far the largest, with over 30,000 samples, even after accounting for only single-patient per-household. As part of our DatasetAdapter module we provide automatic patient selection with the *subsample* parameter, this provides easy exploratory analysis.

The CRELES dataset provides multiple waves, but most signals are missing, the *FeatureFetch* module automatically adjust usable modeling signals, ensuring no time-leakage as target preparation is always estimated using future waves.

The NACC is a restricted access dataset and cognitive scores are not immediately available, however, to enhance this study, our *FeatureFetch* module can be linked with the Alzheimer's Disease Sequencing Project (ADSP), and automatically link results from the Phenotype Harmonization Consortium (PHC) dataset [11] funded by NIA (U24 AG074855, U01 AG068057

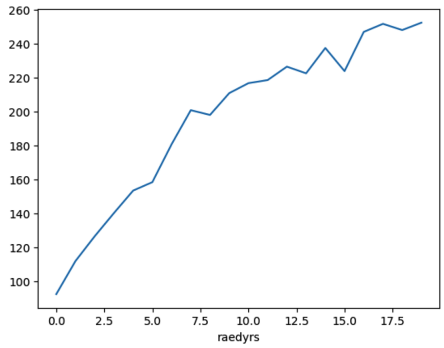

Fig. 2. MHAS: Cognition Score (Y-axis) vs Years of Education

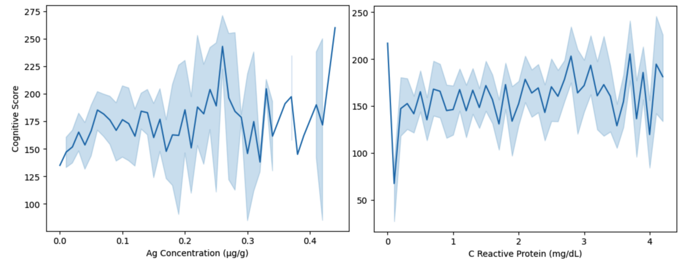

Fig. 3. Top #2 Biomarkers vs observed cognitive scores

and R01 AG059716). We also extended this with access to historical medical records including MRI and PET scans.

All datasets are further harmonized to be processed by the same ADRDModel, this allows the system to operate with new datasets in the future without any changes to the ML model itself, just setting up the right DatasetAdapter configuration.

V. Contributing Factors

In this section we explore key insights that contribute over the entire roadmap of model development.

A. Insight #1: Granular data is better

The original Phase2 generalizes some of the best available features, this includes: Binned Education (*edu_gru_12*), Binned Age (*age_12*) and Reading habits (*reads_12*), but we found the binning strategy to be restrictive. Phase2 data uses education as groups like *"1–5 years"*, same as age with groups like *"60–69"*, using raw values brings better generalization. In Figure 2 we show the relationship between years of education and cognitive scores averages, these signals are lost with binning.

B. Insight #2: Education and correlation traps

Education is one of the most important indicators across studies, showing as Top #1 feature for MHAS, HRS, CRELES and Top #2 for LASI. However, education per-se, might be providing such strong value only in lieu of absent cognitive signals. When removing education features from MHAS base-features, performance can drop as much as 5.5%, but repeating the experiment including historical wave cognitive test-results reduces the effect of removing education-features to only 1%. HRS includes a broader set of features that better represent cognition, hence relying less in education signal, in this dataset, the effect of removing education was 3.4%. It is imperative that we collect wide range of signals which helps the model to avoid over reliance in education, leading to disproportionate mistakes.

C. Insight #3: Biomarkers and recency bias

The modeling of early-prediction usually involves observing at past signals (e.g. previous waves) and estimate a cognitive state or degradation for the future. When all these features are available at all points in time it creates a fair assessment, but in MHAS, we observed different data collections over the years, in 2001 and 2003 anthropometric measures, in 2012 blood samples and extended anthropometric (not seen before), in 2018 hair samples were studied. By picking any arbitrary data into the future, e.g. 2022, a bias will be created giving more recent signals a leak in time, incorrectly increasing its performance. It's important to adjust for this recency bias and analyze individual populations. When assessing biomarker performance removing recency bias none of the collected data provided any strong signal, we studied the two most promising: Concentration of Silver (ag) in hair-samples & C-Reactive Protein (CRP) measuring inflammation and infections as shown in Figure 3, but signals are weak.

D. Insight #4: Historical Cognition Tests

One of the best improvements came from using historical cognition test results. While this could leak the target if not used properly, when exclusively leveraging previous years available tests, it brings real-world improvement, up to 5.5%. MHAS harmonized dataset has multiple waves starting from Wave #1 in 2001 up to Wave #5 in 2018. We studied the effect of how Wave #3 (2012) cognitive tests can serve as predictors for 2018. One must be careful how to interpret the importance of difference tests because they are not all applied at the same rate across waves. To correct this bias, we reviewed within-sample predictive power. Best tests were Visual Scanning (*RwVSCAN*) and Word Recall Summary Score (*RwTR16*). For Visual Scanning respondents were asked to circle all figures that are identical to a specific stimulus shown previously within an array of different stimuli. For Word Recall, it is a summary including both immediate and delayed recall of 8 words each. In Figure 4 we show the predictive power to cognitive score 6-years in the future. These findings might suggest a possible improvement in how a ML dataset is prepared, the original design include patients already with considerable cognitive decline which greats a big bias from the ML standpoint.

E. Insight #5: Feature Selection aiding signal discovery

Studies collect data for a multitude of use-cases which provides the ability to predict multiple targets, but for ADRD prediction, existing datasets tend to bring a lot of unnecessary signals. For MHAS, once all data sources are included, a total of 3384 features are collected. Yet, after feature-importance

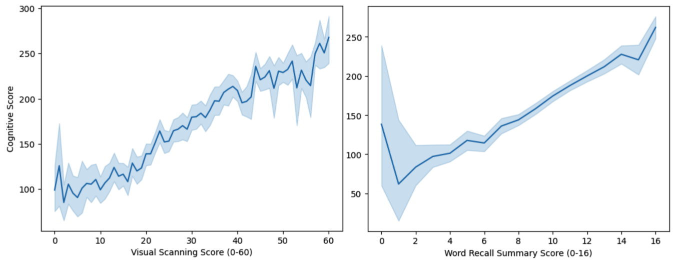

Fig. 4. Top #2 Cognitive Tests vs observed cognitive scores

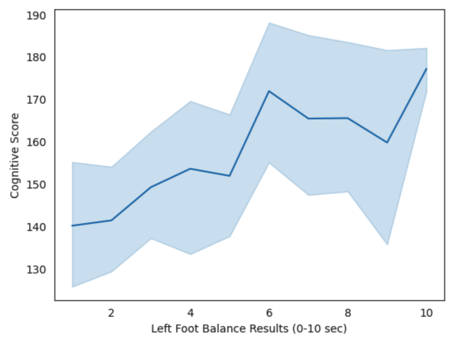

Fig. 5. Left Foot Balance Effect as Cogntion Indicator

analysis, we found 86% of signals can safely be dropped, saving resources and at the same time providing small performance improvements. Once we remove so many redundant or irrelevant signals it is possible to uncover weaker, but still useful candidates like the balance tests (e.g. *RwBALLFSEC*). In Figure 5 we show the effects of left-foot balance, measured in seconds and its relation with cognitive score.

F. Insight #6: Childhood Events Long-term Effect

The HRS dataset provides unique signals dating back to events when candidates were teenagers. In particular *RALHCHILD* counts how many childhood stressful life events the respondent experienced by the age of 16. Incidents include difficult living situation, parents' separation or dead. We studied the individual events and noticed over 70% of value comes from only the parent dead event *RAPADIECH*. This is a strong signal ranking Top #3 only behind education and cognitive tests. This highlights a potential direction for identification of young adults at risk. Among sampled individuals from HRS, when this signal was not missing, up to 30% of candidates suffered this stressful event. In Figure 6 we show the effects, worth notice however is the importance of avoiding study-specific metrics. Note HRS uses z-normalized cognition scores. Greater is better, those with no event average 0.22 z-score while the other group drops to 0.03.

G. Insight #7: Locations as proxy for riskier population

The LASI study with over 35,000 samples makes it possible to do deeper analysis on the influence of residence location and immigration. At first glance, residents from Telangana are much likely to have lower cognitive scores at -0.333σ, compared to regions like Chandigarh with 0.718σ, one standard deviation of difference. This however is a common case of correlation vs causation, on further inspection, Telangana indicates only 42% of its population is illiterate, contrary to Chandigarh with over 85%. After sampling only illiterate population the overall cognitive score goes up. Telangana now 0.382σ and Chandigarh 0.872σ, the difference has reduced in half, but still significant. Illiterate segments are similar age-wise with average of 57.8 & 59.7 for Telangana & Chandigarh respectively. Education shows small deltas with Chandigarh patients having an extra 2 years of education (10 vs 8), we removed the additional education to estimate the effect, sampling candidates from both cities between 5 & 8 years of education only but we kept observing the deltas, Telangana at 0.276σ and Chandigarh 0.699σ. Further findings shown a pattern of immigration, with Telangana population of mostly locals (96%), compared to Chandigarh 23%. Even after controlling for confounding factors there are hidden effects of locations worth exploring, migration patterns seem to have deeper influence than originally thought. When considering only newcomers both states finally look similar with Telangana at 0.56σ and Chandigarh 0.68σ. In Figure 7 we provide final populations after removing effects of age, literacy and education.

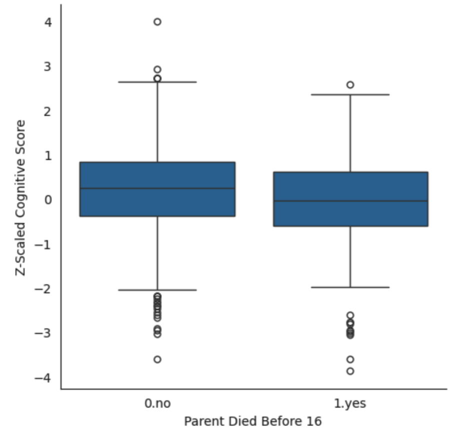

Fig. 6. Observed Effect of Patients Losing a Parent

H. Insight #8: Simpler tests can perform well

The CRELES confirms the heavy importance of education (*raedyrs*) signals as top predictor. However, they include a wide range of test applied to candidates, including: date orientation (*r4orient*), attention for 5-digit backwards count (*r4atten*), instruction-following handling sheet of paper (*r4execu*), delayed word recall (*r4dlrc_cr*), etc. Some tests require much more effort even physical presence and materials (e.g. paper folding), yet, this not always translate to better predictors, the simple date-orientation (e.g. ask day, month, year) shows the best performance. In Figure 8 we provide the relative predictive power of different tests compared to education-signal.

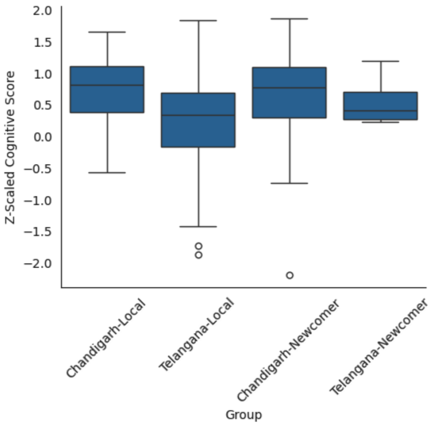

Fig. 7. Location Effect in LASI after confounding removal

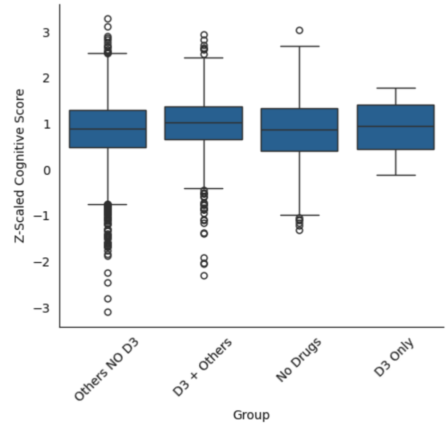

Fig. 9. Cognitive Scores for different D3 interactions

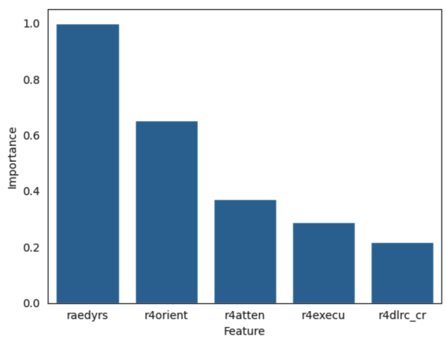

Fig. 8. CRELES: Performance of top tests applied

I. Insight #9: Cholecalciferol suplement improves cognition

The NACC dataset provides great detail of multiple visits from patients, including original diagnosis. We ran a study starting with apparently all-healthy individuals with first-visit diagnosis indicating no concerns on cognitive decline or ADRD risks. We evaluated their journey to understand key differences improving their cognitive functions even in the presence of updated diagnosis. Top indicator relates to Cholecalciferol. From our sample, 2008 patients indicated to be supplementing Vitamin D_3 (*Cholecalciferol*), this represents 22% of population. Since patients can take more than one drug at a time, causing correlations, we further review groups that only take D_3 and compare with those who take no drugs. The result show consistent effect with those taking supplements suffering from lower cognitive decline, both groups where D_3 is supplemented outperform those with or without the presence of other drugs. In Figure 9 we show z-normalized cognitive scores after ADRD diagnosis, largest effect up to 0.17σ between *no-drugs* and *D3+others*.

J. Insight #10: MRI and PET Scans for Alzhimer Detection

We evaluated NACC records for patients with available MRI and PET scans, with 2432 and 1133 cases respectively. The goal is to determine main elements on both imaging techniques that differentiate healthy vs Alzheimer patients. In these studies, the effects of single variables are less significant, and a multivariate analysis is required, we used decisions trees to highlight key feature combinations that identify main differences in patients. For MRI, Left Entorhinal Average Thickness, when smaller than 2.739mm can raise Alzheimer probability by 13%, when doing PET, Right putamen Standardized Uptake Value Ratio (SUVR) greater than 1.464 shows increase risk by 49%. Note that these are not final estimates since a multivariate analysis is required, but more complete details see Figure 10.

VI. Future Direction

Data is the most valuable resource in this field. Although extensive work can be done in the ML model component, the gains are insignificant compared to the value we can achieve with more and cleaner data. Two main directions further exploring include: (1) ML Problem Formulation: we can ask multiple questions to the same data, e.g. "Cognitive Score for a given set of attributes X-years into the future?", similarly to Phase2, a venue we could explore is to change this question, instead, ask "For a set of (at-the-time) perfectly healthy candidates, which ones will develop ADRD in the future?", this is a subtle change but translate to completely different formulation from ML standpoint. Second direction (2) Clinical Signals Exploration: Although not all signals give improvements (e.g. some biomarkers are weak predictors), ML has a big role to play to help discover insights way beyond basic social determinants, advances in ML algorithms can reshape even exiting beliefs for old studies.

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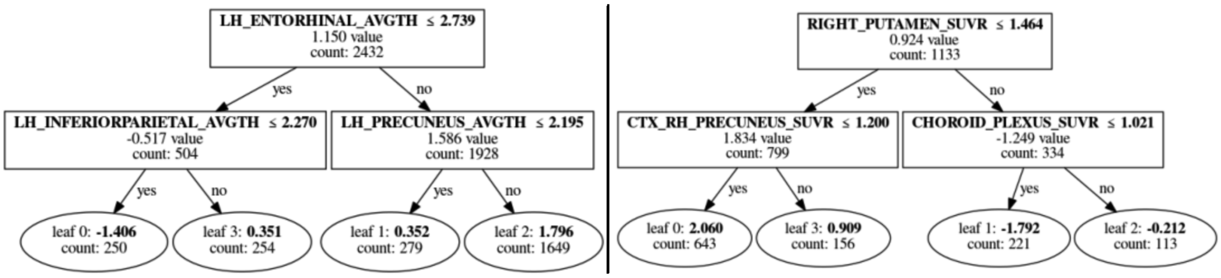

Fig. 10. Decision Tree for Alzhimer Risk in MRI (left) and PET (right)

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